

2025

COMPUTER SCIENCE — MINOR

Paper : MN-3

(Data Structures)

Full Marks : 75

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

1. Answer **any five** questions :

2×5

- (a) Differentiate linear and non-linear data structure.
- (b) Write down two disadvantages of array.
- (c) Define recursion with example.
- (d) Write down two applications of linked list with suitable examples.
- (e) Define an array of five elements and assign (10, 20, 30, 40, 50) values to it.
- (f) What is priority queue? Give an example.
- (g) State any two properties of a binary search tree.
- (h) List any two advantages of doubly linked list over a singly linked list.

Section - A

Answer **any three** questions.

5×3

- 2. What are the differences between stack and queue? Give diagram for both.
- 3. Describe the applications of data structure in real life.
- 4. Convert infix expression into its equivalent postfix expression using stack
 $A * (B + D) / E - F * (G + H / K).$
- 5. Discuss a method to represent a 2D sparse matrix that saves space.
- 6. Differentiate between malloc () and calloc ().

Please Turn Over

(4522)

Section - B

Answer *any five* questions.

7. (a) Write an algorithm to search an element 'X' from 'N' numbers.
(b) Convert the expression into its corresponding prefix expression using stack and evaluate :
 $30 + ((8 - 4) + 20) / 4 + 5 * 3.$ 5+5
8. (a) Represent queue with the help of stack with suitable example.
(b) Explain in detail, how to represent 2D sparse matrix. 5+5
9. (a) Write an algorithm to insert an element in a circular queue.
(b) Perform insertion sort technique on the following set of data in ascending order.
76, 80, 10, 20, 15, 5, 60, 30, 40. 5+5
10. (a) Use hashing technique to store the following set of data and resolve collision if it occurs. Hash function to be used is $\text{key \% } 10$.
90, 12, 23, 38, 56, 69, 77, 83, 86
(b) Write a program snippet to delete a node from a single linked list. Give comments for the program. 5+5
11. (a) Considering linear search and binary search techniques, which technique would you prefer for searching an item and why?
(b) Write the recursive version of the Quick Sort algorithm to sort a given set of elements. Give necessary documentation for the algorithm. 3+7
12. (a) Write the algorithm for the following :
(i) inorder traversal,
(ii) Postorder traversal,
(iii) Preorder traversal of the BST.
(b) What do you mean by a threaded binary tree? Give example. 6+4
13. (a) Write a program in C to reverse the order of elements in a stack using an additional stack. Give necessary documentation for the program.
(b) Implement stack operation using linked list. Provide necessary documentation for the program. 5+5
14. (a) Draw a BST with the following node values :
60, 40, 20, 100, 50, 10, 200, 250, 235, 115, 25
(b) Write the postorder traversal for the above BST.
(c) Delete 200 from the BST and construct the final BST. Show all the steps.
(d) Differentiate between linear and non-linear data structure. 4+2+2+2